

SuperBeam Project Summary

Project started 6 Apr 2019 MS
Site address: 95 Malvern Road
Job: Loft and alterations
Client: Damon
Job number: mal/sdc/01

ITEMS:

1: Beam: Rafters
Span: 1.5 m. Reactions: R1: 0.58 kN R2: 0.58 kN

Use 50 x 125 C16

C16 at 400 c/c

2: Beam: Flat roof joists
Span: 5.3 m. Reactions: R1: 1.86 kN R2: 1.86 kN

Use 50 x 220 C24

C24 at 400 c/c

3: Beam: Dormer Flat roof joists
Span: 2.0 m. Reactions: R1: 0.70 kN R2: 0.70 kN

Use 50 x 125 C24

C24 at 400 c/c

4: Beam: Rafters Back
Span: 3.5 m. Reactions: R1: 1.37 kN R2: 1.37 kN

Use 50 x 170 C16

C16 at 400 c/c

5: Beam: Dormer side rafters
Span: 2.5 m. Reactions: R1: 0.90 kN R2: 1.55 kN

Use 2no 50 x 125 C16

6: Beam: Dormer side rafter supports
Span: 2.0 m. Reactions: R1: 1.56 kN R2: 0.75 kN

Use 2no 50 x 125 C24

7: Beam: Ceiling collars
Span: 1.5 m. Reactions: R1: 0.15 kN R2: 0.15 kN

Use 47 x 125 C24

C24 at 400 c/c

8: Beam: Ridge beam 1
Span: 6.5 m. Reactions: R1: 12.07 kN R2: 11.96 kN
Use 152 x 152 x 44 UC S355
Bearing R1:
Column
Bearing R2: 15 mm m.s. bearing plate, size 300 x 100 mm

9: Beam: Ridge beam 2
Span: 6.5 m. Reactions: R1: 17.95 kN R2: 19.79 kN
Use 203 x 203 x 46 UC S355
Bearing R1:
Column
Bearing R2: 20 mm m.s. bearing plate, size 450 x 100 mm

10: Beam: Ridge beam 3
Span: 7.6 m. Reactions: R1: 10.90 kN R2: 19.46 kN
Use 203 x 203 x 46 UC S355
Bearing R1:
Floor beam
Bearing R2: 20 mm m.s. bearing plate, size 450 x 100 mm

Beam cranks down pitch of hip

11: Beam: T1
Span: 6.6 m. Reactions: R1: 45.53 kN R2: 11.09 kN
Use 203 x 203 x 46 UC S355
Bearing R1: 50 mm m.s. bearing plate, size 1050 x 100 mm
Bearing R2: 10 mm m.s. bearing plate, size 250 x 100 mm

12: Beam: T2
Span: 3.1 m. Reactions: R1: 20.31 kN R2: 15.52 kN
Use 152 x 152 x 23 UC S355
Bearing R1:
To floor beam
Bearing R2: 15 mm m.s. bearing plate, size 350 x 100 mm

13: Steel column: R1 of Ridge beam 2
Unfactored load = 18.0kN
Moments due to load: Mx = 2.87 kNm My = 0.00kNm
Section: 120 x 80 x 10 RHS Grade 50 on 200 x 205 x 10mm base plate
Sum of stress ratios = 0.160 OK

Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10thick to beam via 4 No M16 rbolts

14: Steel column: R1 of Ridge beam1
Unfactored load = 12.1kN
Moments due to load: Mx = 1.93 kNm My = 0.00kNm
Section: 120 x 80 x 10 RHS Grade 50 on 200 x 205 x 10mm base plate
Sum of stress ratios = 0.108 OK

Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10thick to beam via 4 No M16 rbolts

15: Beam: T3
Span: 2.0 m. Reactions: R1: 18.71 kN R2: 7.84 kN
Use 152 x 152 x 23 UC S355
Bearing R1: 20 mm m.s. bearing plate, size 450 x 100 mm
Bearing R2:
Web of T4 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

16: Beam: T4
Span: 4.0 m. Reactions: R1: 4.53 kN R2: 9.38 kN
Use 152 x 89 x 16 UB S355
Bearing R1:
bolts per face.
Bearing R2: 10 mm m.s. bearing plate, size 225 x 100 mm

17: Beam: T5
Span: 4.0 m. Reactions: R1: 4.11 kN R2: 4.87 kN
Use 152 x 89 x 16 UB S355
Bearing R1:
Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
Bearing R2: 10 mm m.s. bearing plate, size 225 x 100 mm

18: Beam: T6
Span: 6.0 m. Reactions: R1: 8.61 kN R2: 5.21 kN
Use 203 x 102 x 23 UB S355
Bearing R1:
Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
Bearing R2:
Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

19: Beam: T8
Span: 6.0 m. Reactions: R1: 7.93 kN R2: 10.53 kN
Use 203 x 133 x 25 UB S355
Bearing R1:
Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
Bearing R2:
Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

20: Beam: T7
Span: 6.0 m. Reactions: R1: 6.55 kN R2: 10.47 kN
Use 203 x 102 x 23 UB S355
Bearing R1:
Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
Bearing R2:
Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

21: Beam: FJ1
Span: 6.0 m. Reactions: R1: 1.79 kN R2: 2.27 kN

Use 75 x 220 C24

22: Beam: FJ2
Span: 6.0 m. Reactions: R1: 1.50 kN R2: 1.50 kN

Use 50 x 220 C24

23: Beam: T9
Span: 1.8 m. Reactions: R1: 2.43 kN R2: 2.43 kN

Use 2no 47 x 220 C24

TITLE

PROPOSED DRAWINGS

ADDRESS

95 MALVERN ROAD
LONDON. NW6

CLIENT

Mr. C. NORMAN

SCALE

1:50 @ A3

DATE

MARCH 2019

DRAWING NO.

190328/30

REV.